KO LIH HAN

Software Engineer | Applied AI & LLM Systems

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EXPERIENCE

Kahap Inc | *Software Developer (Part-time)* Aug 2022 - Aug 2026

- Primary developer for a production feedback platform, building submission workflows, automated notifications, and admin features.
- Maintained a separate Laravel/Vue.js career platform, shipping features and resolving production application and database issues.
- Managed deployments and server maintenance, including migrations/upgrades, Laravel queues/workers, SSL renewal, and database troubleshooting.

Pac Com Solution | *Full-Stack Developer Intern* Jun 2021 - Jan 2022

- Developed and maintained Laravel/Vue.js features for production POS applications.

PROJECTS

Semantic Text-to-SQL | *Python · XiYanSQL 7B · LangGraph · FastAPI · SQLite*

- Built a local Text-to-SQL pipeline using XiYanSQL 7B with schema/value grounding, SQL validation, read-only execution, and bounded repair.
- Reached a 58% execution-match rate on 100 BIRD questions; error analysis showed schema-linking changes fixed 7 cases but broke 7 others.

Local RAG Search | *Python · BM25 · Qwen3-Embedding-0.6B · RRF · FastAPI*

- Built a hybrid retrieval API combining BM25, Qwen3 embeddings, and RRF, with per-result retrieval details for debugging.
- On 64 EnterpriseRAG-Bench queries over 5,189 documents, RRF improved Recall@10 from 0.734 to 0.763 and MRR@20 from 0.679 to 0.742 versus BM25.

Hermes Video Tutor | *Python · Hermes · Qwen3.5 4B · Tool Calling · Multimodal*

- Built a video-QA agent that uses transcript evidence first and only inspects frames or clips when visual information is needed.
- On 12 paired questions, visual tools raised overall pass rate from 16.7% to 50.0%; visual questions reached 66.7%, while transcript-answerable questions avoided visual calls.

SKILLS

AI: RAG, Retrieval, Tool Calling, LangGraph

Programming / Backend: Python, SQL, JavaScript, PHP, FastAPI, REST APIs, Laravel, Vue.js

Data / Infra: MySQL, Redis, SQLite, AWS, Linux, Git, pytest

Languages: Chinese (Native), English (Fluent), Malay (Intermediate)

EDUCATION

National Tsing Hua University | *M.S., Information Systems & Applications* Sep 2024 - Jun 2026

GPA 4.09/4.3 | Taiwan | Thesis: *Interpreting Implicit Intent through Workplace Hierarchies and Context*

Chung Hua University | *B.S., Computer Science & Information Engineering* Sep 2022 - Jun 2024

GPA 3.9/4.0 | Taiwan